ROLL NO: _____

MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

END-SEM EXAMINATION — DECEMBER / JUNE 2023

B.TECH. ECE ENGINEERING — SEMESTER 5

Time Allowed: 3 Hours **Maximum Marks: 70**

NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.
2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].
3. Assume suitable missing data, if any, and state it clearly.
4. Use of scientific non-programmable calculator is permitted.

--- UNIT I ---

Q.1 Explain 8085 Microprocessor Architecture: Registers, ALU, Control unit, and Pin diagram. [14 Marks]

[OR]

Q.1 (Alt) Explain 8085 Instruction cycle: Fetch, Decode, Execute, and Timing diagrams for memory read/write. [14 Marks]

--- UNIT II ---

Q.2 Explain 8086 Microprocessor Architecture: Bus Interface Unit (BIU) and Execution Unit (EU). [14 Marks]

[OR]

Q.2 (Alt) Describe 8086 Memory Segmentation and Physical Address generation. Explain Addressing modes. [14 Marks]

--- UNIT III ---

Q.3 Write 8086 Assembly Language Programs for array sorting and string manipulation. [14 Marks]

[OR]

Q.3 (Alt) Explain 8086 Interrupt structure: Hardware and Software interrupts, Vector Table, and ISR. [14 Marks]

--- UNIT IV ---

Q.4 Explain Programmable Peripheral Interface 8255: Architecture, Modes (0, 1, 2), and BSR mode. [14 Marks]

[OR]

Q.4 (Alt) Describe Interfacing of 8255 with Stepper Motor, Seven Segment Display, and ADC/DAC. [14 Marks]

--- UNIT V ---

Q.5 Explain ARM Cortex-M Architecture: Harvard bus, 3-stage pipeline, and Thumb-2 instructions. [14 Marks]

[OR]

Q.5 (Alt) Compare Microprocessor vs Microcontroller. Explain Embedded C programming and timers. [14 Marks]

**** END OF QUESTION PAPER — MITS GWALIOR AUTONOMOUS EXAMINATION CELL ****